

ROMA, Sweden

WMO: 025950

Lat: 57.5438N

Lon: 18.4253E

Elev: 105

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.6	14.4	4.7	7.0	9.6	10.4	9.4	16.9	26.4	36.9	24.0	36.5	2.3	0	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	79.4	65.6	76.4	63.9	73.7	62.7	68.5	75.2	66.4	72.6	64.6	70.6	8.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.4	70.7	64.3	90.7	68.9	62.3	84.3	67.0	32.8	75.0	31.2	72.7	29.8	70.7	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	20.4	18.2	DB	-1.1	84.8	7.9	3.9	-6.7	87.7	-11.4	90.0	-15.8	92.2	-21.5	95.0	(4)
(5)			WB	-2.2	70.7	8.0	2.4	-8.0	72.4	-12.6	73.9	-17.1	75.2	-22.9	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	29.7	29.8	33.9	41.8	50.5	57.6	63.0	61.9	54.3	45.8	39.7	33.6	(6)	
	DBStd	13.12	7.51	7.34	7.16	5.47	5.43	4.66	4.48	4.53	4.64	5.79	5.60	7.51	(7)	
	HDD50	2998	630	566	499	249	60	4	0	0	16	156	310	508	(8)	
	HDD65	7279	1095	986	964	695	451	226	95	117	323	595	759	973	(9)	
	CDD50	1251	0	0	0	4	75	231	402	370	144	25	0	0	(10)	
	CDD65	56	0	0	0	0	0	3	32	21	0	0	0	0	(11)	
	CDH74	560	0	0	0	0	1	64	328	165	2	0	0	0	(12)	
	CDH80	74	0	0	0	0	0	1	50	23	0	0	0	0	(13)	
	WSAvg	8.7	10.1	9.6	9.2	8.6	8.1	7.8	7.3	7.2	8.3	8.5	9.5	10.0	(14)	
	PrecAvg	21.4	1.8	1.3	1.1	1.0	1.3	1.4	2.2	2.2	2.0	2.1	2.5	2.2	(15)	
(16) Precipitation	PrecMax	28.5	3.7	2.0	2.5	2.2	2.8	4.1	5.4	4.6	4.4	3.8	5.4	4.4	(16)	
	PrecMin	17.0	0.2	0.4	0.2	0.0	0.2	0.0	0.0	0.0	0.6	0.4	0.4	0.6	(17)	
	PrecStd	2.7	0.7	0.5	0.6	0.6	0.7	0.9	1.6	1.2	1.0	0.9	1.0	0.9	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	45.4	55.6	65.5	73.1	79.0	84.5	82.7	73.7	61.8	52.2	48.8	(19)	
		MCWB	43.3	42.2	44.9	50.6	60.3	62.8	67.9	69.5	61.0	55.5	50.3	47.4	(20)	
	2%	DB	42.8	43.0	51.0	60.6	69.5	75.8	80.8	78.1	68.7	58.5	50.3	46.5	(21)	
		MCWB	41.3	40.8	43.7	49.3	57.7	61.7	67.5	65.3	59.5	54.8	48.7	44.8	(22)	
	5%	DB	41.0	40.9	46.5	57.0	65.9	72.8	77.6	75.0	65.8	56.5	48.5	44.5	(23)	
		MCWB	39.4	39.1	41.6	47.6	55.5	60.0	64.9	64.1	58.6	53.7	46.9	43.3	(24)	
	10%	DB	39.2	38.9	43.1	53.4	62.5	69.7	74.6	72.4	63.1	54.7	47.2	42.6	(25)	
		MCWB	37.7	37.4	39.7	44.7	53.4	58.6	63.7	62.9	57.4	52.4	45.8	41.4	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	43.3	46.7	52.6	63.7	66.4	72.6	71.1	63.1	57.6	51.1	47.8	(27)	
		MCDB	44.4	44.3	53.0	61.6	70.5	74.2	79.2	78.8	68.3	59.5	51.7	48.6	(28)	
	2%	WB	41.8	41.3	44.1	50.3	59.8	63.7	69.3	68.2	61.2	55.7	49.0	45.2	(29)	
		MCDB	42.6	42.6	49.5	59.3	66.0	73.0	76.2	74.0	65.7	57.5	49.9	46.2	(30)	
	5%	WB	39.8	39.3	42.2	48.3	57.0	61.5	67.2	66.2	59.6	54.2	47.3	43.4	(31)	
		MCDB	40.6	40.6	46.0	55.5	63.4	69.4	74.2	71.3	64.2	55.9	48.2	44.3	(32)	
	10%	WB	37.8	37.5	40.3	46.1	54.5	59.7	65.4	64.5	58.2	52.8	46.0	41.6	(33)	
		MCDB	38.9	38.8	42.7	51.9	60.8	67.4	71.9	69.5	62.3	54.6	47.0	42.4	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.3	12.5	17.7	18.7	18.6	17.4	17.2	14.4	11.3	7.2	7.0	(35)	
		MCDBR	7.3	7.9	17.9	24.9	24.1	26.1	24.5	22.7	20.0	13.3	7.5	5.7	(36)	
		MCWBR	7.2	6.3	11.6	14.4	13.4	12.9	11.7	11.5	11.7	10.0	6.8	5.8	(37)	
	5% WB	MCDBR	7.2	7.4	16.3	22.3	20.8	22.1	20.6	18.9	17.4	12.0	7.1	5.7	(38)	
		MCWBR	7.2	6.2	11.1	13.5	13.0	12.3	11.1	10.4	10.9	9.6	6.7	5.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.300	0.329	0.343	0.379	0.374	0.365	0.396	0.388	0.357	0.346	0.327	0.292	(40)	
	taud		2.268	2.318	2.361	2.319	2.409	2.473	2.406	2.418	2.479	2.402	2.271	2.200	(41)	
	Ebn at Noon		177	225	256	264	274	278	265	259	251	217	161	140	(42)	
	Edh at Noon		16	23	28	35	34	32	34	31	25	21	16	12	(43)	
(44) All-Sky Solar Radiation	RadAvg		105	283	763	1366	1840	1997	1829	1447	962	430	146	74	(44)	
	RadStd		17	52	75	139	132	91	146	115	107	67	27	13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-196	-254	N/A	N/A

Nomenclature: See separate page